

# ERG Analysis API v2.4.0

## Tier 3 — T3-D: Intended Use & Limitations Statement

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Document ID         | T3-D_Intended_Use_Statement_v1_0                                                                          |
| Product             | ERG Analysis API v2.4.0                                                                                   |
| Organisation        | AI Fairness CIO · Registered Charity No. 1218464 · London, United Kingdom                                 |
| Website             | ai-fairness.com                                                                                           |
| Pipeline file       | erg_v2_4_0.py (Google Colab notebook; Flask server port 8080)                                             |
| Report generator    | erg_report_generator.py (ReportLab PDF)                                                                   |
| Normative ref       | Baker et al. (2025). Doc Ophthalmol 150(2):47–64. DOI: 10.1007/s10633-025-10009-2. N=407.                 |
| ISCEV standard      | Robson et al. (2022). Doc Ophthalmol 144(3):165–177. McCulloch et al. (2015). Doc Ophthalmol 130(1):1–12. |
| OSF pre-reg         | <a href="https://doi.org/10.17605/OSF.IO/6WA42">https://doi.org/10.17605/OSF.IO/6WA42</a>                 |
| SaMD classification | Decision-support software; not a standalone diagnostic device                                             |
| Author              | Hamid Tavakoli, Founder & CIO, AI Fairness CIO                                                            |
| Date                | 24 June 2026                                                                                              |
| Status              | DRAFT — pending Hamid Tavakoli sign-off                                                                   |

### 1. Intended Use

The ERG Analysis API v2.4.0 is a software tool designed to provide automated signal processing, waveform parameter extraction, normative Z-score classification, and clinical decision support for full-field electroretinogram (ERG) recordings obtained in accordance with ISCEV 2022 standards.

#### Intended Use Statement — ERG Analysis API v2.4.0

The ERG Analysis API v2.4.0 is intended for use by qualified eye care professionals and clinical electrophysiologists as a decision-support tool in the assessment of full-field electroretinogram recordings. It is intended to assist, not replace, clinical judgment.

The software processes a digitised ERG waveform, extracts standard ISCEV parameters (a-wave amplitude and implicit time, b-wave amplitude and implicit time, oscillatory potentials, photopic negative response), computes Z-scores against age-stratified reference data from Baker et al. (2025), and generates a traffic-light classification (Green / Amber / Red) to support identification of potentially abnormal findings.

Output must be reviewed and confirmed by a licensed clinician with access to the patient’s full clinical history, symptoms, and examination findings before any clinical action is taken. The output does not constitute a medical diagnosis.

### 1.1 Patient Population

The pipeline is intended for use with ERG recordings obtained from adult patients, corresponding to the Baker et al. (2025) normative population of 407 healthy subjects aged 18–80+ years. Three age strata are supported:

- Stratum le35: age  $\leq 35$  years
- Stratum 36to59: age 36–59 years
- Stratum ge60: age  $\geq 60$  years

The pipeline is not validated for paediatric patients (under 18 years). When a paediatric age group is selected, a visible disclaimer is displayed in the report and Z-scores are computed against the le35 adult stratum as a provisional fallback only.

### 1.2 Supported Recording Conditions

The pipeline is intended for use with recordings that satisfy the following minimum conditions:

- Sampling rate  $\geq 1000$  Hz (pipeline operates at 2000 Hz)
- Bandwidth 0.3–300 Hz (ISCEV 2022 standard bandpass)
- Pre-stimulus baseline  $\geq 20$  ms
- Post-stimulus recording  $\geq 300$  ms
- Electrode type: Gold Foil or DTL Fiber (silver thread) for Z-score classification; Contact Lens and Skin electrodes for raw parameter extraction only
- Protocol: DA 0.01, DA 3, DA 10, LA 3, or LA 30 Hz (ISCEV 2022 standard protocols)
- Flash duration  $< 5$  ms (ISCEV 2022 standard stimulus)

### 1.3 Intended Clinical Setting

The pipeline is intended for use in clinical electrophysiology departments, ophthalmology outpatient settings, and research environments where full-field ERG recordings are obtained under ISCEV-compliant conditions. It is not intended for home use, point-of-care screening without clinical oversight, or emergency settings.

### 1.4 Output Interpretation Guide

The pipeline produces a four-layer report. Clinicians should interpret the outputs as follows:

| Layer   | Output                 | Clinical meaning                                                                                               | Clinician action                                                                                                                                                                                                                          |
|---------|------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 1 | Traffic Light<br>GREEN | All ERG parameters within Baker 2025 age-matched normal limits ( $ Z  \leq 2.0$ for all parameters)            | No immediate electrophysiological action required based on these parameters alone. A GREEN result does not exclude significant ocular or systemic pathology. Integrate with full clinical history, examination, and other investigations. |
| Layer 1 | Traffic Light<br>AMBER | At least one parameter is borderline abnormal ( $2.0 <  Z  \leq 3.0$ , or LA 30 Hz b_amp $< 20$ $\mu$ V floor) | Specialist review recommended. Identify the worst parameter from the Layer 2 metrics table. Consider repeat recording if signal quality is Grade C or D. Do not use as the sole basis for clinical decision-making.                       |
| Layer 1 | Traffic Light<br>RED   | At least one parameter is significantly abnormal ( $ Z  > 3.0$ )                                               | Urgent specialist review required. Identify the worst parameter and its direction (reduction / delay). Integrate with full clinical context. Refer as clinically indicated.                                                               |

| Layer          | Output                     | Clinical meaning                                                                                                               | Clinician action                                                                                                                                                                                                                                                                              |
|----------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Layer 2</b> | <i>Z-score table</i>       | Individual Z-scores for a-wave, b-wave, b/a ratio, and b-wave implicit time against Baker 2025 age-matched norms               | Use to identify which parameter(s) are driving the traffic light signal. A-wave reduction suggests outer retinal (photoreceptor) dysfunction; b-wave reduction with preserved a-wave suggests inner retinal (ON-bipolar) dysfunction. Implicit time delay suggests slowed retinal conduction. |
| <b>Layer 3</b> | <i>Quality grade (A–D)</i> | Signal quality assessment: SNR, OP availability, bandwidth, spike count, ISCEV compliance                                      | Grade A or B: Z-scores are reliable. Grade C: interpret Z-scores with caution; consider repeat. Grade D: Z-scores unreliable; repeat recording required.                                                                                                                                      |
| <b>Layer 4</b> | <i>Technical audit</i>     | Full pipeline parameters, filter log, DTL correction flags, flash duration compliance, inversion polarity flag, version string | For technical review and audit trail only. Not for routine clinical use. Review if the traffic light result is unexpected or if a DTL correction flag is present.                                                                                                                             |

## 2. Contraindications

The pipeline must not be used in the following circumstances:

| #         | Contraindication                                                                  | Rationale                                                                                                                                                                                         | Guidance                                                                                                                                                                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C1</b> | <b>Paediatric patients (under 18)</b>                                             | Baker 2025 normative data covers adults only. Paediatric ERG amplitude and implicit time norms differ substantially from adult values.                                                            | Do not use Z-score traffic light output to guide clinical decisions for patients under 18. A visible disclaimer block is displayed in the report; the pipeline may still be used to process the signal and extract raw parameters for review against laboratory-specific paediatric norms. |
| <b>C2</b> | <b>Real-time / intraoperative monitoring</b>                                      | Zero-phase offline filtering is incompatible with real-time use.                                                                                                                                  | Do not use for intraoperative ERG monitoring or any application requiring real-time output.                                                                                                                                                                                                |
| <b>C3</b> | <b>Standalone diagnosis</b>                                                       | The pipeline is a decision-support tool only.                                                                                                                                                     | Do not use the traffic light output as the sole basis for any clinical decision, diagnosis, or treatment plan. All outputs must be reviewed by a qualified clinician.                                                                                                                      |
| <b>C4</b> | <b>Non-ISCEV recording conditions</b>                                             | The pipeline validates ISCEV 2022 compliance at runtime. Recordings that fail the ISCEV compliance check (sampling rate, bandwidth, pre-stimulus, post-stimulus) may produce unreliable Z-scores. | If iscev_compliance.overall_compliant = False in the Layer 4 audit output, Z-scores and traffic light classifications should be treated as provisional only.                                                                                                                               |
| <b>C5</b> | <b>Contact lens or skin electrode recordings requiring Z-score classification</b> | No Baker 2025 normative data exists for these electrode types in the current schema.                                                                                                              | Z-score classification will not be computed. Raw parameters are reported. Select Gold Foil or DTL for Z-score output.                                                                                                                                                                      |

### 3. Known Limitations

The following limitations are inherent to the current version of the pipeline. They are documented here for regulatory transparency and to guide appropriate clinical use. Each limitation corresponds to a finding in the Tier 3 audit documentation (T3-A through T3-C).

| #  | Category                                 | Limitation                                                                                                                                                                                                                                                                                                                                 | Implication                                                                                                                                                                                                                | Mitigation / Current State                                                                                                                                                                                                                                                   |
|----|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1 | <b>Normative dataset scope</b>           | Baker et al. (2025) N=407 adults, three age strata ( $\leq 35y$ / $36-59y$ / $\geq 60y$ ). No paediatric, sex-stratified, or ethnicity-stratified reference ranges.                                                                                                                                                                        | Z-scores and traffic light classifications are not validated for patients outside the adult 18–80+ year range covered by Baker 2025, for paediatric patients, or for populations not represented in the Baker 2025 cohort. | Paediatric age group inputs trigger a visible disclaimer block in the PDF report and a WARNING flag in the Layer 4 audit output. The pipeline should not be used as a decision-support tool for paediatric patients without laboratory-specific paediatric reference ranges. |
| L2 | <b>Electrode scope</b>                   | Z-score classification is available only for Gold Foil and DTL Fiber (silver thread) electrodes. Contact lens and skin electrodes produce raw parameter output only.                                                                                                                                                                       | Clinicians using contact lens or skin electrodes cannot use the traffic light classification. Raw amplitude and implicit time values are still reported.                                                                   | The report displays "NORMATIVE UNAVAILABLE" with an explanation when an unsupported electrode is selected. Selection of Gold Foil or DTL is required for Z-score classification.                                                                                             |
| L3 | <b>Protocol scope</b>                    | Five ISCEV standard protocols are supported: DA 0.01, DA 3, DA 10, LA 3, LA 30 Hz. No multifocal ERG, pattern ERG, or extended protocol variants.                                                                                                                                                                                          | The pipeline cannot be used for mfERG, PERG, or non-standard protocol recordings.                                                                                                                                          | Protocol selection is explicit at input. Unsupported protocols are not silently processed; the pipeline will either reject the input or produce no Z-scores.                                                                                                                 |
| L4 | <b>Training data basis</b>               | The ERG Analysis API v2.4.0 uses signal processing rules and normative Z-scores derived from Baker et al. (2025). The Stage 1 / Stage 2 machine learning classifiers described in the OSF pre-registration (DOI: 10.17605/OSF.IO/6WA42) are not yet trained or deployed. Stage 2 disease classification output is NOT available in v2.4.0. | The pipeline cannot classify disease type (DR, glaucoma, AMD, retinitis pigmentosa / rod dystrophy, cone dystrophy, complete CSNB). Traffic light output reflects Z-score deviation from normative data only.              | Layer 3 Specialist output is clearly labelled PENDING in the report. The OSF pre-registration records aspirational AUC targets (Stage 1 $\geq 0.94$ , Stage 2 macro-AUC $\geq 0.80$ ) as pre-registered goals, not achieved results.                                         |
| L5 | <b>DTL Deming correction uncertainty</b> | Deming regression slopes for DTL-to-Gold-Foil amplitude conversion range from 0.46 to 0.73, with $r^2$ values of 0.68–0.85. The LA 3 a_amp                                                                                                                                                                                                 | DTL Z-scores carry higher uncertainty than Gold Foil Z-scores, particularly for LA 3 a-wave. The inversion                                                                                                                 | The _la3_a_amp_dtl_r2_borderline flag is set in the Layer 4 audit output when LA 3 a_amp is computed for a DTL electrode. All Deming correction parameters are documented in the T3-B Normative Traceability Matrix.                                                         |

| #  | Category                 | Limitation                                                                                                                                                                                                                                                                     | Implication                                                                                                                                                                                              | Mitigation / Current State                                                                                                                                                                                                                                                      |
|----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                          | conversion has a borderline $r^2=0.68$ .                                                                                                                                                                                                                                       | formula amplifies measurement noise by a factor of $1/\text{slope}$ ( $1.4-2.2\times$ ). Borderline Amber results on DTL should be interpreted with additional caution.                                  |                                                                                                                                                                                                                                                                                 |
| L6 | Gaussian normative model | Z-scores assume a Gaussian distribution of ERG parameters within each Baker 2025 stratum. The lower Green bound ( $\mu - 2\sigma$ ) for some ge60 amplitude strata approaches physiologically implausible values (e.g. LA 30 Hz b_amp ge60 lower Green = 29.8 $\mu\text{V}$ ). | Very low but Gaussian-within-normal amplitude values in elderly patients could receive a GREEN classification despite clinical significance.                                                             | An absolute lower-bound floor of 20 $\mu\text{V}$ is applied for LA 30 Hz b_amp: any value below this threshold overrides to AMBER regardless of Z-score (T3-C B4, implemented in v2.4.0). Additional absolute floors for other protocols are a recommended future enhancement. |
| L7 | Single-sweep input       | The pipeline accepts a single averaged ERG sweep as input. It does not implement averaging across multiple sweeps, artefact rejection across a sweep set, or inter-sweep variability analysis.                                                                                 | Single-sweep recordings are more susceptible to noise, artefact, and trial-to-trial variability than averages of multiple sweeps. Quality Grade D recordings should be treated with particular caution.  | The ERG signal quality audit (Layer 3) assigns a grade A–D and reports SNR, spike count, saturation, EMG level, and ISCEV compliance. Clinicians should repeat recordings for Grade C or D results.                                                                             |
| L8 | Offline processing only  | The filtering pipeline uses zero-phase Butterworth filtering (sosfiltfilt), which requires the full signal to be available. Real-time / streaming processing is explicitly blocked (apply_streaming_filter() raises NotImplementedError).                                      | The pipeline cannot be used for real-time intraoperative ERG monitoring or online feedback systems.                                                                                                      | Streaming processing is scheduled for a future version. The NotImplementedError prevents accidental causal-filter use.                                                                                                                                                          |
| L9 | No diagnostic conclusion | The pipeline produces a Z-score-based traffic light classification and, in future versions, a probabilistic disease classification. It does not produce a diagnostic conclusion, a clinical recommendation, or a treatment plan.                                               | The output of the pipeline is decision support only. It must be reviewed and interpreted by a licensed clinician with access to the patient's full clinical history, symptoms, and examination findings. | Every report carries a mandatory disclaimer: "This report is generated by an artificial intelligence pipeline and is intended solely as a decision-support tool for qualified eye care professionals. It does not constitute a medical diagnosis."                              |

## 4. Relationship to OSF Pre-registration

The ERG Analysis API v2.4.0 is governed by a pre-registered Statistical Analysis Plan (OSF DOI: 10.17605/OSF.IO/6WA42). The following clarifications apply to the relationship between the pre-registration and the current implementation:

| Pre-registration item                                                            | Current v2.4.0 status                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stage 1 binary classifier (Normal / Abnormal; AUC <math>\geq 0.94</math>)</b> | NOT YET IMPLEMENTED. The AUC $\geq 0.94$ target is a pre-registered aspirational goal, not an achieved result. v2.4.0 uses Z-score-based traffic light classification in lieu of the trained classifier. Layer 3 Specialist output is labelled PENDING.                   |
| <b>Stage 2 six-class disease classifier (macro-AUC <math>\geq 0.80</math>)</b>   | NOT YET IMPLEMENTED. The six disease classes (DR, Glaucoma, AMD, Retinitis Pigmentosa / Rod Dystrophy, Cone Dystrophy, Complete CSNB) and the $\geq 0.80$ macro-AUC target are pre-registered aspirational goals. No disease classification output is produced in v2.4.0. |
| <b>Total output classes = 7 (1 Normal + 6 disease)</b>                           | Architecture is pre-registered and locked. Implementation is deferred to Tier 4 / external validation phase.                                                                                                                                                              |
| <b>Normative reference: Baker et al. (2025)</b>                                  | IMPLEMENTED. All Z-scores in v2.4.0 are computed against Baker et al. (2025) N=407, DOI: 10.1007/s10633-025-10009-2, via the locked normative JSON (schema_version 1.0).                                                                                                  |
| <b>DTL Deming regression correction</b>                                          | IMPLEMENTED. Baker 2025 Table 2 coefficients loaded from normative JSON. Inversion formula applied to DTL amplitude parameters only. Peak times not transformed.                                                                                                          |
| <b>FHIR R4 output (LOINC 70943-7)</b>                                            | IMPLEMENTED. FHIR Observation resource generated for every run, including LOINC 70943-7 top-level code, SNOMED CT interpretation codes, and custom ERG parameter component codes.                                                                                         |

## 5. Regulatory Context

### 5.1 Software as a Medical Device (SaMD) Classification

The ERG Analysis API v2.4.0 is software that processes physiological measurement data and generates outputs intended to inform clinical decisions. Under the MHRA AI/ML SaMD framework and the EU MDR / UKCA framework, it is likely to be classified as a Class I or Class IIa SaMD (decision-support tool, non-standalone diagnostic).

The pipeline does not produce a standalone diagnosis, does not automatically trigger clinical action, and explicitly requires clinician review of all outputs. These design features are consistent with a decision-support classification.

### 5.2 Compliance with ISCEV 2022

The pipeline is designed to comply with the ISCEV Standard for full-field clinical electroretinography (Robson et al., 2022; McCulloch et al., 2015). Compliance is enforced and documented as follows:

- Recording parameters (sampling rate, bandwidth, pre-stimulus, post-stimulus) are validated at runtime and written to the Layer 4 ISCEV compliance audit block.
- Notch filter is OFF by default. User activation requires explicit consent and is logged with a warning in the filter log.

- Flash duration is validated against the 5 ms ISCEV maximum; non-compliance is flagged in the Layer 4 audit.
- Protocol-specific measurement windows for all five standard ISCEV protocols are implemented and locked.
- Formal ISCEV compliance audit findings are documented in T3-A\_ISCEV\_Compliance\_Checklist\_v1\_0.

### 5.3 AI Fairness CIO and BiasClean™

The ERG Analysis API is developed under the AI Fairness CIO (Registered Charity No. 1218464) charitable mission of promoting fair, transparent, and bias-reduced AI in clinical applications. The pipeline implements the BiasClean™ framework principles:

- Stratified reference norms: Baker 2025 three-stratum age framework applied universally.
- Explicit uncertainty quantification: Z-scores with locked SD-based thresholds; DTL borderline  $r^2$  flagged.
- Transparent audit trail: Layer 4 technical audit and FHIR output record all processing decisions.
- Explicit limitation disclosure: paediatric, sex, and ethnicity coverage gaps are documented and flagged in the report.

Bias and fairness audit findings are documented in T3-C\_Bias\_Fairness\_Audit\_v1\_0.

## 6. Mandatory Report Disclaimer

The following disclaimer appears on every PDF clinical report generated by the pipeline and in the Layer 2 clinical summary JSON output. It must not be removed or modified.

#### **Mandatory Disclaimer — reproduced verbatim from `erg_report_generator.py`**

"This result is generated by an automated signal analysis algorithm and is intended to support, not replace, clinical judgment. It does not constitute a medical diagnosis. All results should be interpreted in the context of the patient's full clinical history, examination findings, and relevant investigations by a qualified healthcare professional. A Green ERG signal does not exclude significant ocular or systemic pathology. If you have any concern about this patient's visual function, please refer to an appropriate specialist."

The Layer 2 clinical summary in the pipeline JSON output also carries an AI-specific disclaimer:

#### **Layer 2 AI disclaimer — reproduced verbatim from `erg_v2_4_0.py`**

"This report is generated by an artificial intelligence pipeline and is intended solely as a decision-support tool for qualified eye care professionals. It does not constitute a medical diagnosis and must not be used as the sole basis for clinical decision-making. All findings must be reviewed and confirmed by a licensed clinician with access to the patient's full clinical history, symptoms, and examination findings in accordance with ISCEV 2022 standards."

## 7. Sign-off

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This document is classified DRAFT. It will be reclassified APPROVED upon counter-signature by Hamid Tavakoli.

| Role                     | Name                         | Date         |
|--------------------------|------------------------------|--------------|
| T3-D Author (Claude AI)  | AI Fairness CIO ERG Pipeline | 24 June 2026 |
| Founder & CIO (Approval) | Hamid Tavakoli               | Pending      |

End of T3-D Intended Use & Limitations Statement v1.0 — AI Fairness CIO (Reg. No. 1218464)